Check Point 4

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1 Introduction

This analysis uses data from the National Health and Nutrition Examination Survey (NHANES) Vision and Eye Health Surveillance dataset. NHANES is a nationally representative survey of the U.S. civilian, non-institutionalized population. The dataset includes demographic variables (e.g., Sex) and vision-related prevalence estimates (Data_Value) across survey cycles.

2 Objective and approach

2.1 Specific objective

This report examines whether the mean prevalence estimate (Data_Value) for a selected vision-related outcome differs between males and females. Simulations are used to evaluate how effect size, variability (noise), and sample size influence the estimated difference between sexes.

2.2 Data Description

```r
# Packages
library(tidyverse)

# Set working directory
setwd("/Users/lynnliu/Desktop/R Data/")

nhanes_eye <- read_csv(
  "National_Health_and_Nutrition_Examination_Survey_(NHANES)_-_Vision_and_Eye_Health_Surveillance_20260
)

# Data Table

data_description <- data.frame(
  Variable = c("Sex",
               "Data Value",
               "Question",
               "Response",
               "Sample Size"),
  Type_in_Dataset = c("Categorical",
                      "Continuous",
                      "Categorical",
                      "Categorical",
                      "Discrete"),
  Class_in_R = c(class(nhanes_eye$Sex),
                 class(nhanes_eye$Data_Value),
                 class(nhanes_eye$Question),
                 class(nhanes_eye$Response),
                 class(nhanes_eye$Sample_Size))
)

knitr::kable(
  data_description,
  caption = "Description of key variables used in the analysis",
  align = c("l", "c", "c")
)
```

Table 1: Description of key variables used in the analysis

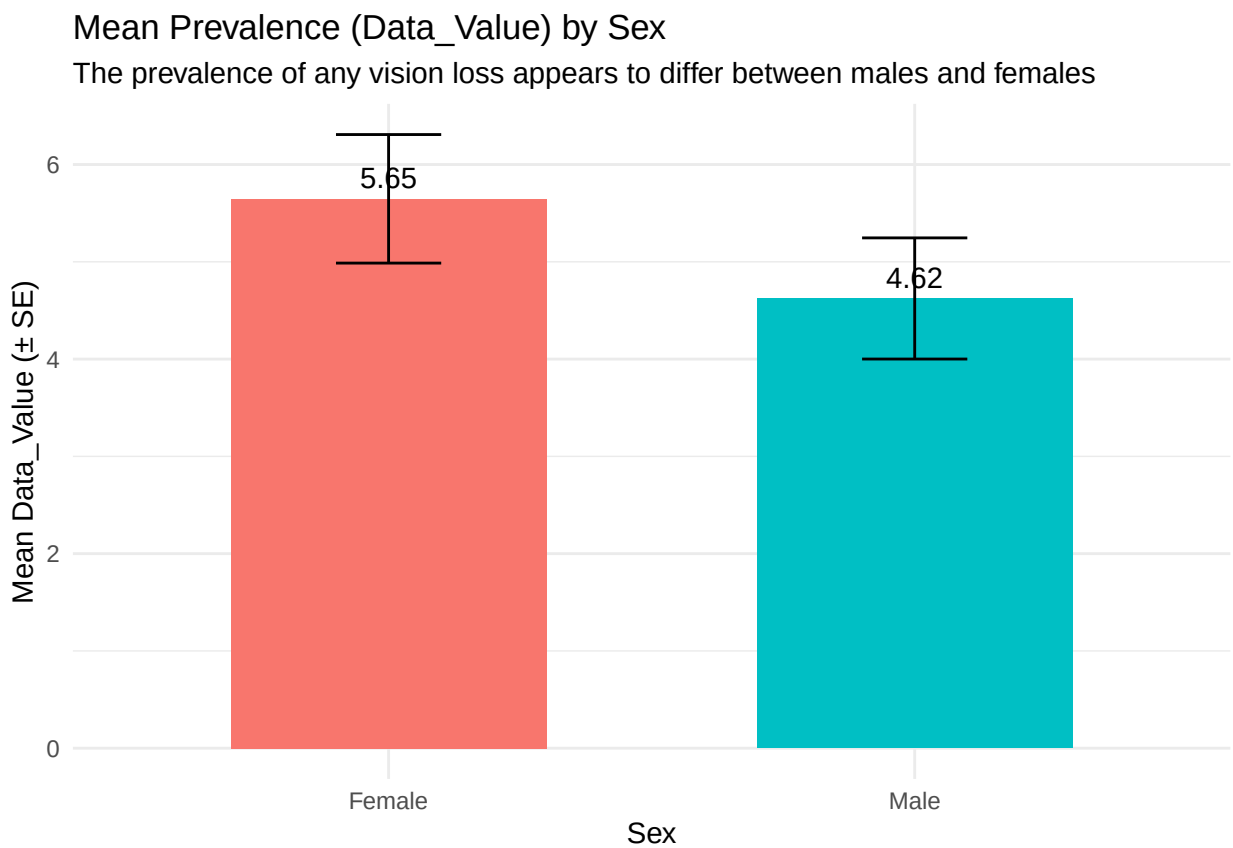
Variable	Type_in_Dataset	Class_in_R
Sex	Categorical	character
Data Value	Continuous	numeric
Question	Categorical	character
Response	Categorical	character
Sample Size	Discrete	numeric

2.3 Data Visualization

```r
# Keep only usable rows
df <- nhanes_eye %>%
  filter(Sex %in% c("Male", "Female")) %>%
```

```r
  filter(!is.na(Data_Value)) %>%
  filter(is.finite(Data_Value))

# Select one Question + Response that has both sexes
pick <- df %>%
  group_by(Question, Response) %>%
  summarise(n_sex = n_distinct(Sex), .groups = "drop") %>%
  filter(n_sex == 2) %>%
  slice(1)

chosen_question <- pick$Question[1]
chosen_response <- pick$Response[1]

analysis_df <- df %>%
  filter(Question == chosen_question,
         Response == chosen_response)

# Summary table
analysis_df %>%
  group_by(Sex) %>%
  summarise(
    Mean_Value = mean(Data_Value),
    SD_Value = sd(Data_Value),
    n = n(),
    .groups = "drop"
  )
```

```r
# Visualization
plot_df <- analysis_df %>%
  group_by(Sex) %>%
  summarise(
    Mean_Value = mean(Data_Value),
    SD_Value = sd(Data_Value),
    n = n(),
    SE = SD_Value / sqrt(n),
    .groups = "drop"
  )

ggplot(plot_df, aes(x = Sex, y = Mean_Value, fill = Sex)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = round(Mean_Value, 2)),
            vjust = -0.5) +
  geom_errorbar(aes(ymin = Mean_Value - SE,
                    ymax = Mean_Value + SE),
                width = 0.2) +
  labs(
    x = "Sex",
    y = "Mean Data_Value (± SE)",
    title = "Mean Prevalence (Data_Value) by Sex",
    subtitle = paste0("The prevalence of any vision loss appears to differ between males and females")
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



2.4 Plauisble Relationships

The visualization suggests that the average Data_Value may differ between males and females. This relationship can be described as a difference in group means, calculated as the mean Data_Value among females minus the mean Data_Value among males. This difference represents the effect size examined in the simulation.

2.5 Parameters of Interest

```
parameters <- data.frame(
  Parameter = c("Baseline mean (males)", "Effect size", "Noise level", "Sample size per group"),
  Symbol = c("mu0", "delta", "sigma", "n"),
  Description = c("Mean prevalence among males",
                  "Female mean minus Male mean",
                  "Standard deviation of random noise in simulation",
                  "Number of observations per sex")
)

knitr::kable(parameters, caption = "Parameters used in the simulation")
```

Table 2: Parameters used in the simulation

Parameter	Symbol	Description
Baseline mean (males)	mu0	Mean prevalence among males
Effect size	delta	Female mean minus Male mean
Noise level	sigma	Standard deviation of random noise in simulation
Sample size per group	n	Number of observations per sex

3 Simulation

3.1 Simulation Basis

```
male_mean <- analysis_df %>%
  filter(Sex == "Male") %>%
  summarise(m = mean(Data_Value)) %>%
  pull(m)

pooled_sd <- analysis_df %>%
  summarise(s = sd(Data_Value)) %>%
  pull(s)
```

3.2 Simulation Output

```
# Example simulation parameters
effect_size <- 0.5
noise_sd <- pooled_sd
n_per_group <- 100
mu0 <- male_mean

# Generate one simulated dataset (A = Data_Value, B = Sex)
sim_data <- tibble(
  Sex = rep(c("Male", "Female"), each = n_per_group),
  Data_Value = c(
    rnorm(n_per_group, mean = mu0, sd = noise_sd),
    rnorm(n_per_group, mean = mu0 + effect_size, sd = noise_sd)
  )
)

# Stratified summary statistics data frame (quartiles of A stratified by B)
summary_stats <- sim_data %>%
  group_by(Sex) %>%
  summarise(
    n = n(),
    mean = mean(Data_Value),
    sd = sd(Data_Value),
    q1 = quantile(Data_Value, 0.25),
    median = quantile(Data_Value, 0.50),
    q3 = quantile(Data_Value, 0.75),
```

```r
    .groups = "drop"
  )
```

```r
# Display the summary statistics data frame
knitr::kable(
  summary_stats,
  caption = "Stratified summary statistics of simulated Data Value by Sex"
)
```

Table 3: Stratified summary statistics of simulated Data Value by Sex

Sex	n	mean	sd	q1	median	q3
Female	100	4.127958	6.381296	-0.0683773	4.159174	8.267967
Male	100	4.831662	5.493347	0.6864968	4.914385	7.339027

3.3 Simulation Function

```r
#3.3 Simulation Function

run_simulation <- function(effect_size, noise_sd, n_per_group, mu0) {

  sim_data <- tibble(
    Sex = rep(c("Male", "Female"), each = n_per_group),
    Data_Value = c(
      rnorm(n_per_group, mean = mu0, sd = noise_sd),
      rnorm(n_per_group, mean = mu0 + effect_size, sd = noise_sd)
    )
  )

  summary_stats <- sim_data %>%
    group_by(Sex) %>%
    summarise(
      n = n(),
      mean = mean(Data_Value),
      sd = sd(Data_Value),
      q1 = quantile(Data_Value, 0.25),
      median = quantile(Data_Value, 0.50),
      q3 = quantile(Data_Value, 0.75),
      .groups = "drop"
    )

  female_mean <- summary_stats$mean[summary_stats$Sex == "Female"]
  male_mean   <- summary_stats$mean[summary_stats$Sex == "Male"]
  estimated_difference <- female_mean - male_mean

  list(
    data = sim_data,
    summary_stats = summary_stats,
    estimated_difference = estimated_difference
```

```
  )
}
```

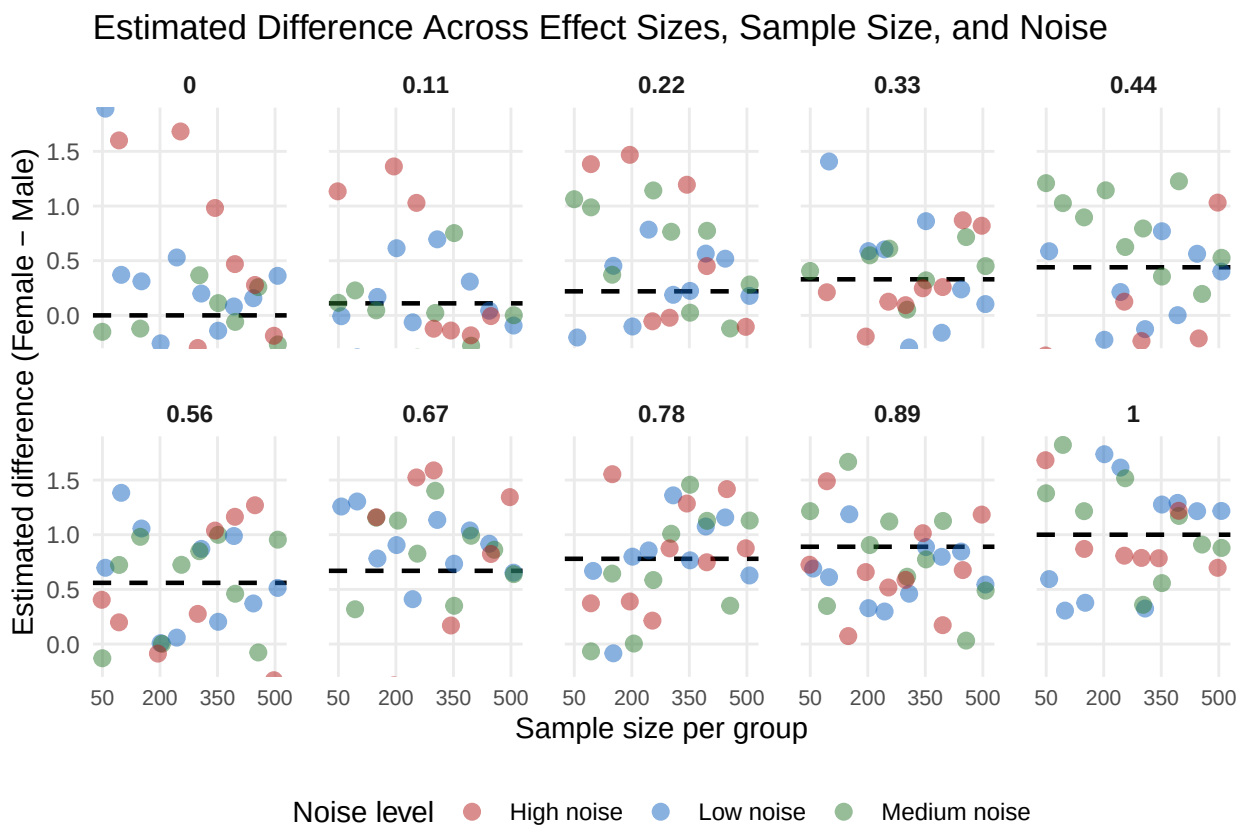
3.4 Simulation Automation

```
effect_sizes <- seq(0, 1, length.out = 10)
sample_sizes <- round(seq(50, 500, length.out = 10))
noise_levels <- pooled_sd * c(0.75, 1, 1.5)

simulation_results <- list()
counter <- 1

for (e in effect_sizes) {
  for (n in sample_sizes) {
    for (noise in noise_levels) {

      res <- run_simulation(
        effect_size = e,
        noise_sd = noise,
        n_per_group = n,
        mu0 = male_mean
      )

      simulation_results[[counter]] <- list(
        effect_size = e,
        n_per_group = n,
        noise_sd = noise,
        data = res$data,
        summary_stats = res$summary_stats,
        estimated_difference = res$estimated_difference
      )

      counter <- counter + 1
    }
  }
}
```

4 Visualization

```
# Build summary dataframe from simulation results
sim_summary <- purrr::map_dfr(simulation_results, ~ tibble(
  effect_size = .x$effect_size,
  n_per_group = .x$n_per_group,
  noise_sd = .x$noise_sd,
  estimated_difference = .x$estimated_difference
)) %>%
  mutate(
    effect_size = round(effect_size, 2),
    n_per_group = factor(n_per_group),
```

```r
    noise_lab = case_when(
      noise_sd == min(noise_levels) ~ "Low noise",
      noise_sd == median(noise_levels) ~ "Medium noise",
      TRUE ~ "High noise"
    )
  )

# Keep x-axis labels readable
x_breaks <- levels(sim_summary$n_per_group)[seq(1, length(levels(sim_summary$n_per_group)), by = 2)]

# Final visualization
ggplot(sim_summary,
       aes(x = as.numeric(as.character(n_per_group)),  # ← numeric, not factor
           y = estimated_difference,
           color = noise_lab)) +
  geom_hline(aes(yintercept = effect_size), linetype = "dashed",
             linewidth = 0.8, color = "black") +
  geom_point(
    alpha = 0.5,
    size = 2.5,
    position = position_jitter(width = 8, height = 0)  # ← jitter in actual units
  ) +
  facet_wrap(~ effect_size, ncol = 5) +
  scale_x_continuous(
    breaks = c(50, 200, 350, 500),  # ← only show a few clean labels
    labels = c("50", "200", "350", "500")
  ) +
  scale_color_manual(values = c(
    "Low noise"    = "#1565C0",
    "Medium noise" = "#2E7D32",
    "High noise"   = "#B71C1C"
  )) +
  labs(
    color = "Noise level",
    x = "Sample size per group",
    y = "Estimated difference (Female - Male)",
    title = "Estimated Difference Across Effect Sizes, Sample Size, and Noise"
  ) +
  coord_cartesian(ylim = c(-0.2, 1.8)) +
  theme_minimal(base_size = 11) +
  theme(
    legend.position = "bottom",
    axis.text.x = element_text(size = 8),
    panel.spacing = unit(1.1, "lines"),
    strip.text = element_text(size = 9, face = "bold"),
    panel.grid.minor = element_blank()
  )
```



5 Interpretation

These simulations show that larger effect sizes (greater differences in mean Data_Value between females and males) produce more stable estimated differences across repeated samples. When the true effect size is small, larger sample sizes are needed to obtain consistent estimates. Higher noise increases variability in the estimated differences, especially at smaller sample sizes.

6 AI Use Disclosure Statement

This document was generated using ChatGPT to assist with understanding the assignment instructions and drafting initial R code for data subsetting, visualization, and probability calculations.